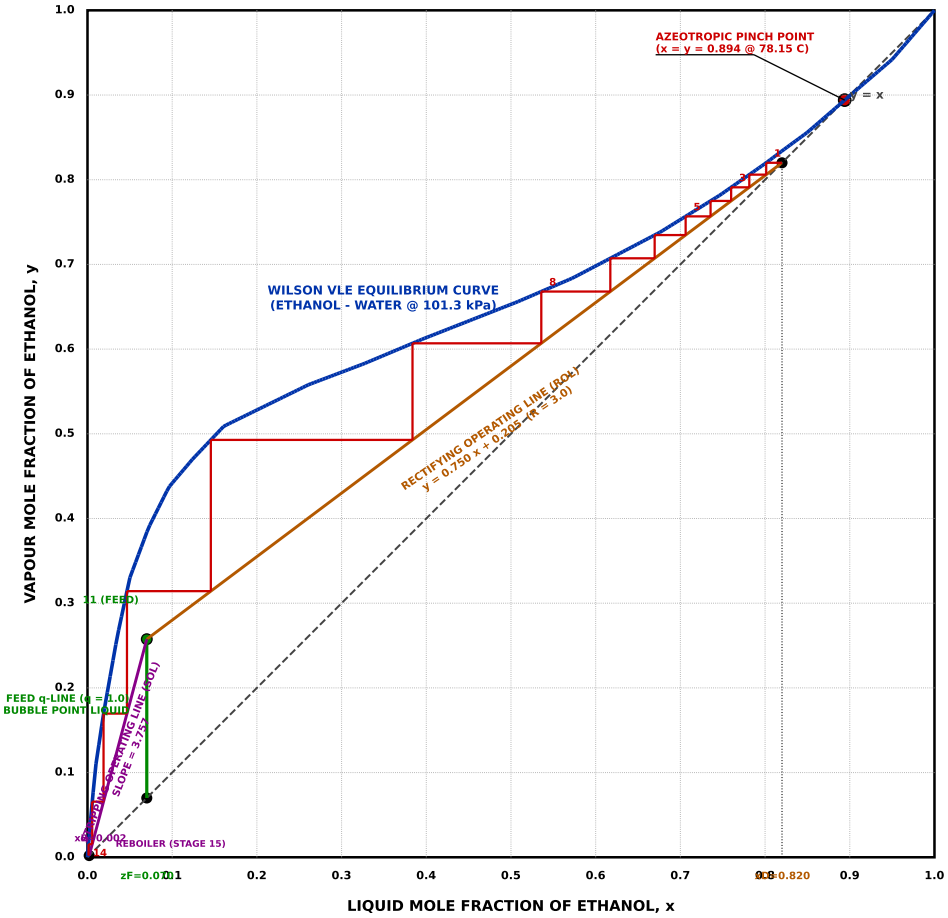
VAPOUR-LIQUID EQUILIBRIUM (WILSON MODEL) & McCABE-THIELE STAGE DETERMINATION

SYSTEM: ETHANOL(1) - WATER(2) @ 101.3 kPa | R = 3.0 | FEED: zF = 0.070, xD = 0.820, xB = 0.002



THERMODYNAMIC BASIS: WILSON ACTIVITY COEFFICIENT MODEL

ln(gamma_1) = -ln(x_1 + L12*x_2) + x_2 * [L12/(x_1 + L12*x_2) - L21/(L21*x_1 + x_2)]

ln(gamma_2) = -ln(x_2 + L21*x_1) - x_1 * [L12/(x_1 + L12*x_2) - L21/(L21*x_1 + x_2)]

PARAMETERS: L12 = 0.1664 | L21 = 0.8856 | V1 = 58.68 cm3/mol | V2 = 18.07 cm3/mol

AZEOTROPIC BEHAVIOR: Minimum-Boiling Azeotrope @ 78.15 C, x_azeo = 0.8943 mole fraction

ANTOINE (EtOH): log10(P) = 8.20417 - 1642.89 / (T + 230.3) [P in mmHg, T in C]

TABLE 1: McCABE-THIELE COLUMN SIZING & STAGE SPECIFICATIONS

PARAMETER	VALUE	UNIT	DESIGN BASIS / REMARKS
Feed Flowrate (F)	700.0	kmol/h	Dilute ethanol wash from fermenter (13,985 kg/h)
Feed Composition (zF)	0.070 (7.00)	mol%	Equivalent to 16.14 wt% ethanol in water
Distillate Comp. (xD)	0.820 (82.00)	mol%	High grade industrial ethanol (91.90 wt%)
Bottoms Comp. (xB)	0.002 (0.20)	mol%	Stripped effluent (97.38 % ethanol recovery)
Minimum Reflux Ratio (Rmin)	1.93	-	Calculated via tangent pinch analysis @ x=0.75
Operating Reflux Ratio (R)	3.00	-	R / Rmin = 1.55 (Optimal economic design ratio)
Boilup Ratio (S = Vb/B)	2.40	-	Provides required stripping reboiler duty
Theoretical Stages (N_theor)	14 Trays + Reb	stages	15 Total equilibrium stages via McCabe-Thiele
Optimal Feed Stage	Stage 11	-	Rectifying: 10 stages | Stripping: 4 stages + Reb
Murphree Tray Efficiency (Eo)	65.0	%	O Connell correlation for aqueous alcohol sieve tray
Actual Number of Trays	24 Trays	trays	N_theor / Eo = 15 / 0.65 = 23.1 -> 24 Sieve Trays
Column Internal Diameter (Dc)	1.20 (1200)	m (mm)	Sized for 68.3 % flooding @ max vapor load
Tray Spacing (Ts)	450	mm	Standard industrial single-pass crossflow sieve tray

TABLE 2: HYDRAULIC OPERATING WINDOW & FLOODING VALIDATION

HYDRAULIC METRIC	RECTIFYING	STRIPPING	LIMIT / STANDARD
Max Vapor Flow, V (kg/h)	9,548	9,910	Sized at column top & reboiler return
Max Liquid Flow, L (kg/h)	7,161	21,146	Includes internal reflux + liquid feed
Gas Velocity, uv (m/s)	1.12	1.18	Design operating superficial velocity
Flooding Velocity, uf (m/s)	1.64	1.69	Fair flooding correlation (Csb = 0.088)
Percent Flooding (%)	68.3 %	69.8 %	Allowable: 65% - 75% max for stability
Total Tray DP (Pa / tray)	540 Pa	580 Pa	Well within 650 Pa design limit
Downcomer Backup (mm liq)	165 mm	182 mm	< 50% tray spacing (225 mm limit)
Weeping Check (h_crest, mm)	14.2 mm	18.5 mm	> 10 mm (Zero weeping operation)

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
VLE ANALYSIS (WILSON MODEL) & McCABE-THIELE STAGES	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	DIAGRAMMATIC / GRAPHICAL
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/ED/DWG-02	01 / 09 / 2026

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA